

Arc, sector and the net of a cone

Cambridge insert: the radian formulas of Grade 10, used to build a cone from a flat sheet.

LESSON 38

1 × 40 MIN

GEOMETRY 11, §17 (EXTENSION)

P1 · 4.2–4.3

LESSON CLOCK

10'

14'

12'

6'

3'

The link between the two circles	10 min
Finding the angle	14 min
Waste	12 min
A design task	6 min
Homework	3 min

LEARNING OBJECTIVES

- Use $s = r\theta$ and $A = \frac{1}{2}r^2\theta$ in a practical construction.
- Find the sector angle needed to build a cone of given dimensions.
- Compute the wasted material when a sector is cut from a sheet.
- Design a cone to a specification.

TEXTBOOK

National	pp. 191–194
Cambridge	Pure Mathematics 1, pp. 86–95
Workbook	P1 Exercise 4B

01

Explanation

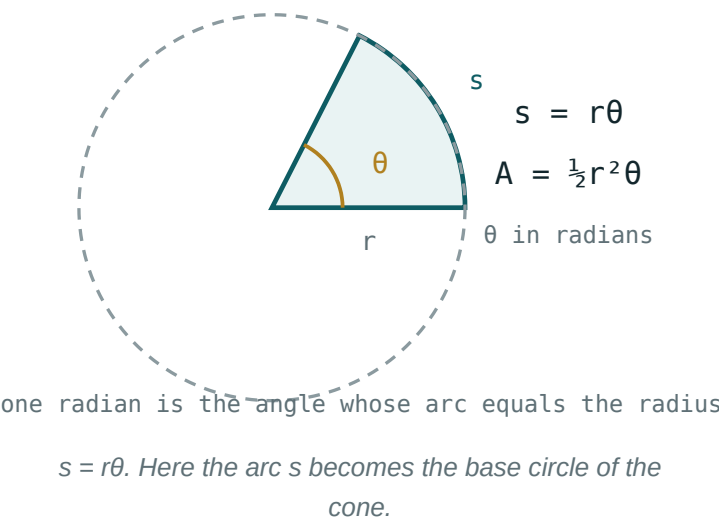
The link between the two circles

THE ONE EQUATION

The arc of the sector becomes the circumference of the base:

$$l\theta = 2\pi r$$

with θ in radians, l the sector radius (the cone's slant height) and r the cone's base radius.



Everything else follows: $\theta = \frac{2\pi r}{l}$ in radians, or $\frac{360r}{l}$ degrees.

Finding the angle

L	R	Θ	IN DEGREES
12	4	$\frac{2\pi}{3}$	120°
10	5	π	180°
15	6	$\frac{4\pi}{5}$	144°
l	l	2π	360°

The last row is the flat disc: when $r = l$ the cone has no height at all.

Check the areas agree: the sector area $\frac{1}{2}l^2\theta = \frac{1}{2}l^2 \cdot \frac{2\pi r}{l} = \pi rl$ — the curved surface area of the last lesson, derived a second way.

Waste, and a design task

A sector is cut from a square or a circular sheet; the rest is waste. The proportion wasted is a real design consideration:

waste fraction from a full disc = $1 - \frac{\theta}{2\pi} = 1 - \frac{r}{l}$

Example. A cone with $r = 4$, $l = 12$ uses $\frac{1}{3}$ of the disc of radius 12, so two thirds of the material is wasted. A shallower cone — r closer to l — wastes far less.

THE DESIGN TASK, IN PAIRS

Design a paper cone to hold exactly 200 ml (200 cm^3). Choose r and h , compute l and θ , draw the sector, cut and build it. Then measure what it actually holds and explain the difference.

02 Worked examples

EXAMPLE 1 A cone has $r = 5$ and $l = 15$. Find the sector angle in degrees.

$$\textcircled{1} \quad \theta = \frac{2\pi r}{l} = \frac{10\pi}{15} = \frac{2\pi}{3}$$

$$\textcircled{2} \quad \text{In degrees: } \frac{360 \times 5}{15}$$

$$\textcircled{3} \quad = 120^\circ$$

ANSWER

120°

EXAMPLE 2 A sector of radius 20 and angle 216° is rolled up. Find the cone's radius and volume.

$$① \quad r = \frac{216}{360} \times 20 = 12$$

$$② \quad h = \sqrt{400 - 144} = 16$$

$$③ \quad V = \frac{1}{3} \pi (144)(16)$$

$$④ \quad = 768\pi \approx 2413$$

ANSWER

$$r = 12, V = 768\pi \approx 2413$$

EXAMPLE 3 Design a cone of volume 200 cm^3 with $r = 5$. Find h , l and θ .

$$① \quad 200 = \frac{1}{3} \pi (25)h \Rightarrow h = \frac{24}{\pi} \approx 7.64$$

$$② \quad l = \sqrt{25 + 58.3} \approx 9.13$$

$$③ \quad \theta = \frac{360 \times 5}{9.13} \approx 197^\circ$$

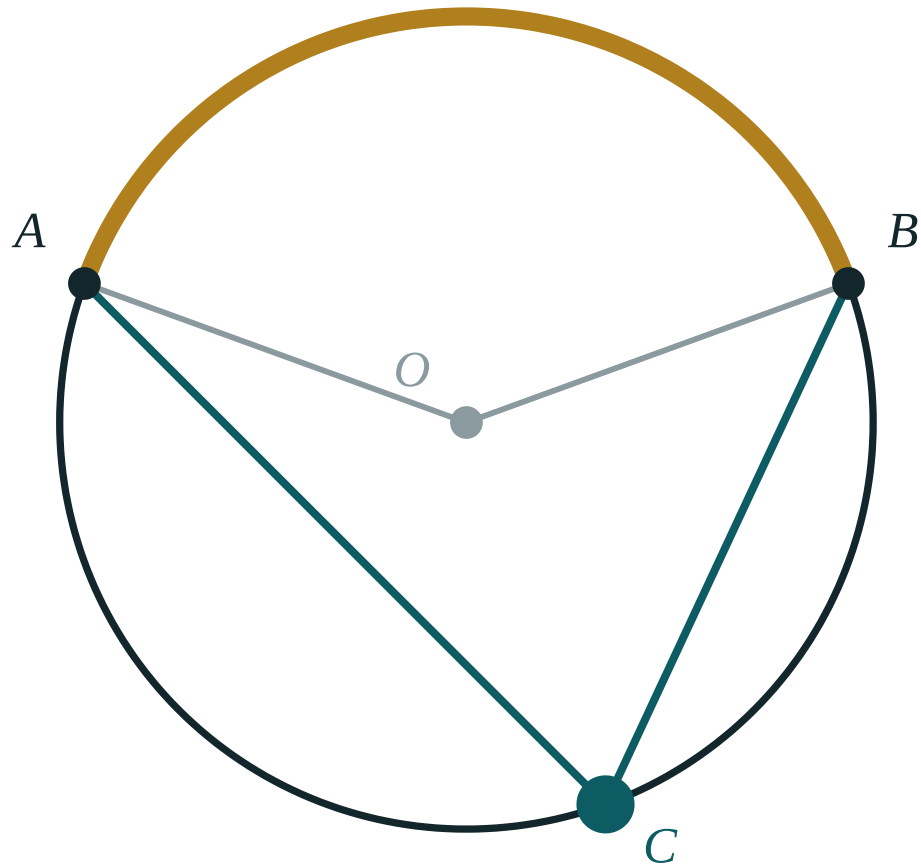
ANSWER

$$h \approx 7.6, l \approx 9.1, \theta \approx 197^\circ$$

03 Interactive model

Cut two sectors of the same radius but different angles and compare the cones.

Sector to cone

central $\angle AOB$ **140°** inscribed $\angle ACB$ **70°** ratio **2**arc AB **140°** **04 Terminology**

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Sector	Sektor	Сектор
Arc length	Yoy uzunligi	Длина дуги
Sector angle	Sektor burchagi	Угол сектора
Radian	Radian	Радян
Net	Yoyma	Развёртка
Waste material	Chiqindi material	Отходы материала
Circumference	Aylana uzunligi	Длина окружности
Design specification	Loyiha talabi	Техническое задание

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The arc of the sector becomes:

the slant height

the base circumference

the height

the axis

θ in radians is:

$$\frac{r}{l}$$

$$\frac{2\pi r}{l}$$

$$\frac{l}{r}$$

$$2\pi l$$

$l = 12, r = 4$ gives θ :

$$60^\circ$$

$$120^\circ$$

$$180^\circ$$

$$240^\circ$$

When $r = l$ the cone is:

very tall

a flat disc

a sphere

impossible

The sector area equals:

πr^2

$\pi r l$

$2\pi r l$

πl^2

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. θ for $l = 10, r = 5$

ANSWER 180°

2. θ for $l = 12, r = 4$

ANSWER 120°

3. θ for $l = 15, r = 6$

ANSWER 144°

4. r from $l = 20, \theta = 90^\circ$

ANSWER 5

5. r from $l = 18, \theta = 180^\circ$

ANSWER 9

6. The arc becomes

ANSWER the base circumference

7. Sector area equals

ANSWER $\pi r l$

Medium · the standard question

7 PROBLEMS

- Sector radius 20, angle 216° : the cone's radius

ANSWER 12

- Same: the height and volume

ANSWER 16, 768π

- Cone of $r = 5$, $l = 15$: the angle

ANSWER 120°

- Waste fraction for $r = 4$, $l = 12$

ANSWER $\frac{2}{3}$

- Waste fraction for $r = 9$, $l = 10$

ANSWER 0.1

- Cone of volume 200 with $r = 5$: h

ANSWER ≈ 7.64

- Same: the sector angle

ANSWER $\approx 197^\circ$

Hard · stretch and reason

7 PROBLEMS

1. A cone of maximum volume from a disc of radius R : the sector angle

ANSWER $\approx 294^\circ$

2. That maximum volume for $R = 12$

ANSWER ≈ 836

3. A cone with $\theta = 90^\circ$ and volume 200: find l

ANSWER ≈ 11.7

4. Two cones from one disc of radius 10, no waste: their radii

ANSWER 5 each if the disc is halved

5. A cone of $r = 6$, $h = 8$: the sector angle and the waste from a disc of radius l

ANSWER $216^\circ, 40\%$

6. Design a cone of volume 500 and $\theta = 180^\circ$

ANSWER $r \approx 6.2, l \approx 12.4$

7. Why does a shallower cone waste less material?

ANSWER r is nearer l , so the sector is nearer the full disc

07 Homework

Homework — 4 tasks

Bring the built paper cone and its measured capacity.

1. A cone has $r = 8$ and $l = 20$. Find the sector angle in degrees and in radians.
2. A sector of radius 24 and angle 150° is rolled up. Find the cone's radius, height and volume.
3. Design and build a paper cone of volume 200 cm^3 , giving r , h , l and θ .
4. Measure what your cone actually holds and explain in two sentences why it differs from 200 ml.